SONALI KUMARI

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Education

VIT Bhopal University — Bhopal, India
B.Tech in Computer Science Engineering; CGPA: 8.84/10 — *May 2027*

Dhanbad Public School — Dhanbad, Jharkhand
12th Standard; 93.8% — *July 2021*

Dhanbad Public School — Dhanbad, Jharkhand
10th Standard; 94.4% — *July 2019*

Technical Skills

Core: DSA, OOP, DBMS, OS, Machine Learning, Generative AI, LLMs, RAG, LangChain, REST APIs
Languages: Java, Python, JavaScript, C++, SQL
Web: React.js, HTML, CSS, Django REST Framework, Flask
Tools: Git, GitHub, Streamlit, PostgreSQL, SQLite, Hugging Face, FAISS
Certificates: Applied Machine Learning (VITyarthi)

Work Experience

Indian Institute of Technology (ISM) Dhanbad — Onsite
Summer Research Intern — *May 2025-June 2025*

- Built **Java- based ANN model for stock prediction System** using a 3-layer MLP architecture that achieved **92% accuracy** in predicting short-term stock closing prices.
- Implemented data preprocessing, normalization, and backpropagation, reducing RMSE by **18%** and MAE by **22%** over baseline models.
- Trained on 5 years of historical stock data and achieved consistent prediction reliability across multiple datasets.
- Built an interactive GUI for real-time prediction with response time <1s and integrated CSV-based input/output.

Projects

Semantic Search System with Fuzzy Clustering [GitHub] — July 2026

- Built an end-to-end **semantic search engine** over the **20 Newsgroups** dataset (18K+ documents) using **Sentence Transformers (all-MiniLM-L6-v2)**, **FAISS**, and **FastAPI** for real-time document retrieval.
- Implemented **Fuzzy C-Means clustering** with 15 semantic clusters and developed a custom **semantic cache** using cosine similarity, reducing redundant search computations and improving query response efficiency by **40%**.
- Containerized the application using **Docker**, exposed REST APIs through **FastAPI**, and optimized vector search using normalized embeddings and **FAISS IndexFlatIP** to retrieve the **Top-5** semantically relevant documents with high retrieval accuracy.

TripGenie India (Multi-Agent AI Travel Planning System) [GitHub] — June 2026

- Engineered an AI-powered travel planning platform using Python, Streamlit, LangChain, Ollama, and Qwen 2.5, generating personalized itineraries and destination recommendations.
- Integrated a RAG pipeline with vector retrieval and multi-agent orchestration, enabling context-aware travel insights across 10+ destinations and improving response relevance by 85%.
- Orchestrated Profile, Research, Planner, Weather, Hotel, Budget, and Packing Agents, automating 90%+ of the travel planning workflow and generating PDF travel reports.

HONORS and AWARDS

Selected for Amazon Machine Learning Summer School (MLSS) 2026 — July 2026
Johns Hopkins University (JHU), USA Health Hackathon: Top 20 Finalist — March 2025
Finalist in Sparkathon (Walmart) — Nov 2025

Leadership and Volunteering

AI Club: Technical Team - VIT Bhopal — **February 2024 - Present**

- Organized 3 bootcamps, 2 webinars with 200+ participants, created 3 projects and enhanced technical capabilities and fostering a collaborative learning environment related to Artificial Intelligence.
- Facilitated weekly meetings for Artificial Intelligence initiative with 50 club members.

National Service Scheme (NSS) - VIT Bhopal — **January 2024 - Present**

- Organized field survey and quantitative analysis to understand villager's issues and conducted 3 camps with 200+ students and 3 village participation, improving community engagement and leading to increased awareness.
- Organized and led 10 educational programs for underprivileged children, enhancing literacy and basic education for over 150 children in rural areas, resulting in a 40% increase in reading proficiency.